



**NATIONAL
SENIOR CERTIFICATE**

GRADE 11

NOVEMBER 2023

**AGRICULTURAL SCIENCES P2
MARKING GUIDELINE**

MARKS: 150

This marking guideline consists of 9 pages.

SECTION A**QUESTION 1**

1.1	1.1.1	C ✓✓		
	1.1.2	A ✓✓		
	1.1.3	A ✓✓		
	1.1.4	C ✓✓		
	1.1.5	B ✓✓		
	1.1.6	C ✓✓		
	1.1.7	B ✓✓		
	1.1.8	D ✓✓		
	1.1.9	D ✓✓		
	1.1.10	B ✓✓	(10 x 2)	(20)
1.2	1.2.1	F ✓✓		
	1.2.2	D ✓✓		
	1.2.3	A ✓✓		
	1.2.4	J ✓✓		
	1.2.5	C ✓✓	(5 x 2)	(10)
1.3	1.3.1	Transpirational pull ✓✓		
	1.3.2	Agricultural lime ✓✓		
	1.3.3	Integrated Pest Control ✓✓		
	1.3.4	Biotechnology ✓✓		
	1.3.5	Soil surveying ✓✓	(5 x 2)	(10)
1.4	1.4.1	Fertilisation ✓		
	1.4.2	Oxygen ✓		
	1.4.3	Monoculture ✓		
	1.4.4	Legumes ✓		
	1.4.5	Fallow ✓	(5 x 1)	(5)

TOTAL SECTION A: 45

SECTION B**QUESTION 2: PLANT STUDIES (NUTRITION)**

- 2.1 2.1.1 **Process shown in the diagram**
Photosynthesis ✓ (1)
- 2.1.2 **Identification of product**
Sugar ✓ (1)
- 2.1.3 **Storage organs for products of photosynthesis**
• Roots ✓
• Stems ✓
• Leaves ✓
• Seeds ✓ (Any 2 x 1) (2)
- 2.1.4 **Factors influencing the rate of photosynthesis**
• Light intensity ✓
• Temperature ✓
• Carbon dioxide concentration ✓
• Water ✓ (Any 2 x 1) (2)
- 2.1.5 **Methods of optimising photosynthesis**
• Use of a greenhouse / tunnel ✓
• Plant density ✓
• Trellising plants ✓
• Pruning ✓ (Any 2 x 1) (2)
- 2.1.6 **Consequences of the absence of photosynthesis**
• Plants and animals will die due to lack of oxygen ✓
• Animals will die of hunger ✓
• Global warming ✓ due to absence of plants that absorb carbon dioxide ✓ (Any 2 x 1) (2)
- 2.2 2.2.1 **Identification of processes A and B**
A – Osmosis ✓
B – Transpiration ✓ (2)
- 2.2.2 **Importance of water movement in plants**
• Transport of nutrients ✓
• Temperature regulation ✓
• Uptake of nutrients ✓ (Any 2 x 1) (2)
- 2.2.3 **Ways plants use to reduce water loss through transpiration**
• Waxy cuticle ✓
• Hairy leaves ✓
• Small leaves ✓
• Stomata on underside of leaves ✓ (Any 2 x 1) (2)

- 2.3 2.3.1 **Labels**
 B – Micro ✓
 C – Potassium ✓
 D – K⁺ ions ✓
 E – NO₃⁻ / NH₄⁺ ✓
 F – Stunted growth / Leaf chlorosis ✓
 G – Macro ✓ (6 x 1) (6)

- 2.3.2 **Methods that can be used to determine the nutrient status of soils**
 • Soil analysis ✓
 • Plant analysis ✓ (2)

- 2.4 2.4.1 **Difference between passive and active mineral uptake**

Passive uptake	Active uptake
• Nutrients move down the concentration gradient ✓	• Nutrients move against the concentration gradient ✓
• No energy required ✓	• Energy in the form of ATP required ✓
• Carrier molecule not required ✓	• Carrier molecule required ✓

(Any 4 x 1) (4)

- 2.4.2 **Factors affecting nutrient availability**
 Soil texture ✓
 Soil pH ✓ (2)

- 2.5 2.5.1 **Classification of fertilisers**
 Farmer A – Organic fertiliser ✓
 Farmer B – Inorganic fertiliser ✓ (2)

- 2.5.2 **Environmental effects of inorganic fertilisers**
 • Can lead to eutrophication ✓
 • Soil acidification ✓
 • Storage and application of nitrogenous fertilisers emits greenhouse gases ✓
 • Can destroy soil food webs ✓ (Any 2 x 1) (2)

- 2.5.3 **An example of a nitrogenous fertiliser**
 Lime ammonium nitrate / Ammonium nitrate ✓ (Any 1 x 1) (1)

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QUESTION 3: PLANT REPRODUCTION AND PROTECTION

- 3.1 3.1.1 **Captions for diagrams 1 and 2**
Diagram 1 – Stamen / androecium ✓
Diagram 2 – Pistil / gynoecium ✓ (2)
- 3.1.2 **Function of Part C in diagram 2**
Fuses with the male gamete ✓ (1)
- 3.1.3 **Identification of flower parts**
B – stigma ✓
D – ovary ✓ (2)
- 3.1.4 **Name of process**
Pollination ✓ (1)
- 3.1.5 **Examples of pollination agents**
• Insects ✓
• Birds ✓
• Water ✓
• Wind ✓ (Any 2 x 1) (2)
- 3.1.6 **Letter of a part that will develop into:**
(a) – D ✓
(b) – C ✓ (2)
- 3.2 3.2.1 **Identification of processes**
A – Parthenocarpy ✓
B – Ablactation ✓ (2)
- 3.2.2 **Types of process A**
• Vegetative parthenocarpy ✓
• Stimulative parthenocarpy ✓ (2)
- 3.2.3 **Factors that influence the process B**
• Strong winds ✓
• Low temperatures / frost ✓
• Mineral deficiencies ✓
• Too many fruits ✓
• Insect pests ✓
• Deformed flowers ✓ (Any 2 x 1) (2)
- 3.3 **Identification of the process**
- 3.3.1 Germination ✓ (1)
- 3.3.2 **Name of the process**
Dormancy ✓ (1)

- 3.3.3 **Strategies for overcoming dormancy**
• Priming / soaking seeds ✓
• Scarification ✓ (Any 2 x 1) (2)
- 3.3.4 **Requirements for seed germination**
• Favourable temperature
• Adequate water supply
• Availability of oxygen (Any 1 x 1) (1)
- 3.4 3.4.1 **Name of parts A and B**
A – Scion ✓
B – Root stock ✓ (2)
- 3.4.2 **Classification of reproduction type**
Asexual reproduction ✓ (1)
- 3.4.3 **Justification of answer to QUESTION 3.4.2**
Does not involve the fusion of gametes ✓ (1)
- 3.4.4 **Advantages of asexual reproduction**
• Fertilisation does not need to take place ✓
• Avoid overcrowding plants ✓
• Root stock and scion can be from different varieties ✓
• Can be used where seed propagation is not practical ✓
• Produces fruits that are true to type ✓
• When introducing new cultivars it not necessary to remove the old trees and replace them with young trees ✓ (Any 2 x 1) (2)
- 3.5 3.5.1 **Identification of the role of weeds in plant disease outbreaks**
They serve as hosts for pathogens ✓ (1)
- 3.5.2 **Examples of mechanical weed control methods**
• Fire ✓
• Cultivation ✓
• Mowing / Slashing / cutting ✓ (Any 2 x 1) (2)
- 3.5.3 **Contributions of the state towards plant protection**
• Quarantine services ✓
• Seed certification systems ✓
• Legislation ✓
• Research ✓
• Depositories ✓ (Any 3 x 1) (3)
- 3.5.4 **Examples of stored grain pests**
• Rice / Maize weevil ✓
• Grain beetle ✓
• Grain / Flour moth ✓
• Flour beetle ✓ (Any 2 x 1) (2)

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QUESTION 4: OPTIMAL RESOURCE UTILISATION**4.1 4.1.1 Identification of farming system**

Precision farming ✓

(1)

4.1.2 Key technologies used in precision farming

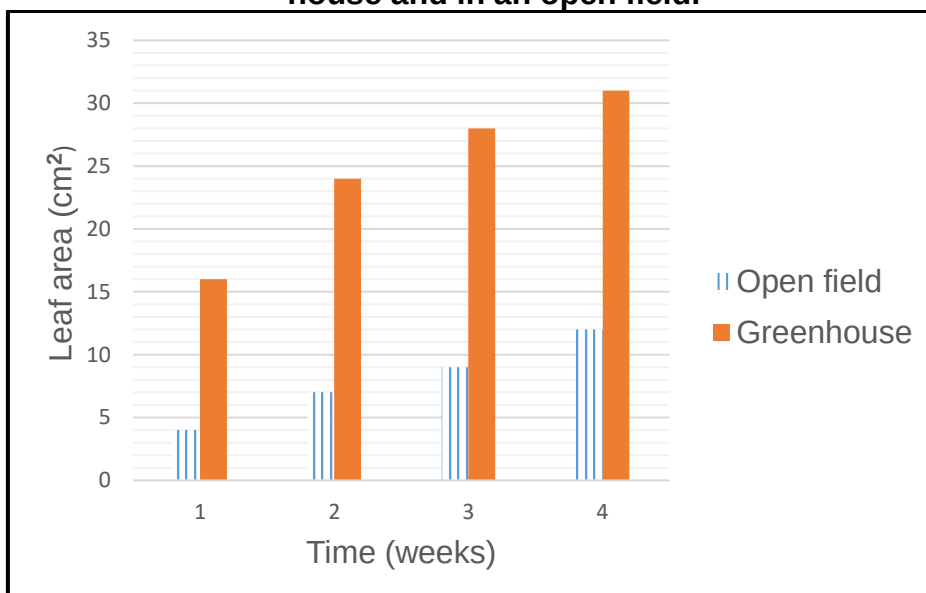
- Global Positioning System / GPS ✓
- Geographical Information System / GIS ✓

(2)

4.1.3 Aims of precision farming

- To allow the farmer to have more precise control over natural variation in soil on the farm ✓
- To enable the farmer to apply inputs only as needed and where needed to ensure optimum production ✓

(2)

4.2 4.2.1 Comparison between the leaf size of spinach grown in a green house and in an open field.

- Correct heading with both variables ✓
- Correct graph type ✓
- Units (Weeks / cm²) ✓
- y-axis: Correct labelling and calibration ✓
- x-axis: Correct labelling and calibration ✓
- Correct plotting (80% and more correct plotting) ✓

(6)

4.2.2 Conclusion that can be drawn from the information in the graph

Plants grown in greenhouses have bigger leaves ✓ than those grown in an open field over 4 weeks. ✓

(2)

4.2.3 Explanation for the differences in leaf sizes

Plants grown in the green house grow faster because temperatures are higher ✓ in the greenhouse, resulting in higher photosynthetic rates. ✓

(2)

- 4.3 4.3.1 **Aims of soil cultivation**
- Bury plant remains ✓
 - Incorporate fertiliser and lime into the soil ✓
 - Control weeds ✓
 - To break up soil crusts ✓
 - Loosen the soil to make root penetration and infiltration easier ✓
- (Any 2 x 1) (2)
- 4.3.2 **Differences between primary and secondary cultivation**
- Primary tillage cut and shatters the soil with deep penetration tools to produce a rough surface finish ✓ whereas secondary tillage aims to level and firm the top part of the soil to produce a smoother surface finish. ✓ (2)
- 4.3.3 **Tools used in primary and secondary cultivation**
- Primary cultivation**
- Ploughs ✓
 - Rippers ✓
- Secondary cultivation**
- Harrows ✓
 - Cultivators ✓
- (Any 2 x 1) (2)
- 4.3.4 **Environmentally friendly cultivation systems**
- Minimum tillage ✓
 - Zero tillage ✓
 - Mulching method ✓
- (Any 2 x 1) (2)
- 4.4 4.4.1 **Identification of process**
- Crop rotation ✓ (1)
- 4.4.2 **Principles of crop rotation shown in the table**
- Crops that require the same nutrients should not follow each other ✓
 - Crops that are affected by the same pests and diseases should not follow each other ✓
 - Shallow rooted crops should be followed by deep rooted crops ✓
- (Any 2 x 1) (2)
- 4.4.3 **Reasons for recommending crop rotation**
- Protects the farmer from total crop failure ✓
 - Maintains soil fertility ✓
 - Improves soil structure ✓
 - Controls pests and diseases ✓
 - Increases soil nitrogen content ✓
- (Any 2 x 1) (2)
- 4.5 4.5.1 **Identification of cultivation system**
- Hydroponics ✓ (1)

4.5.2 Difference between *hydroponics* and *open field system*

Hydroponics is involves growing plants in a water-based nutrient solution ✓ while the open field involves growing crops directly onto the soil. ✓

(2)

4.5.3 Examples of growth media that can be used in hydroponics

Vermiculite ✓

Perlite ✓

(2)

4.5.4 Advantages of the closed hydroponic system

- Uniform irrigation ✓
- Savings on water and fertiliser ✓
- No growth medium is necessary ✓

(Any 2 x 1)

(2)

[35]**TOTAL SECTION B: 105****GRAND TOTAL: 150**